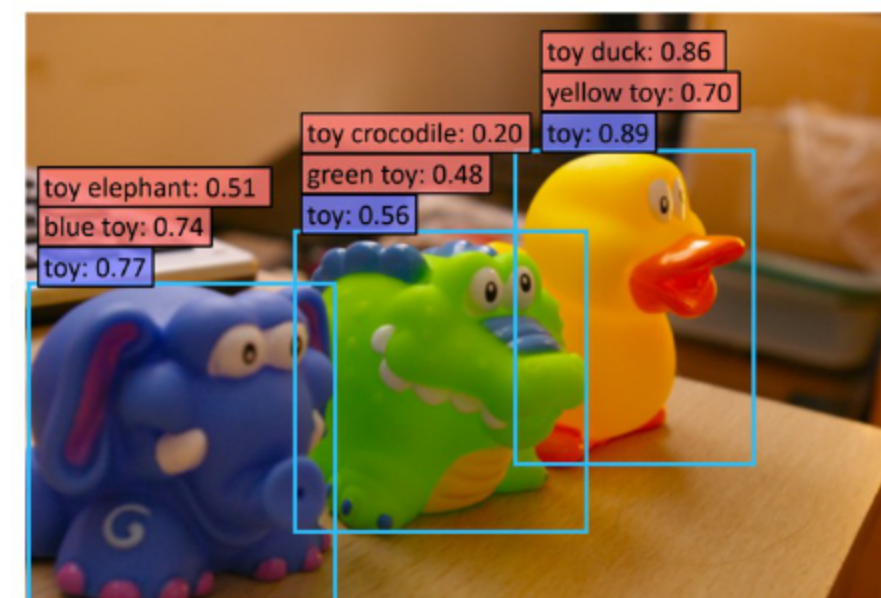


ViLD: Open-Vocabulary Detection via Vision-Language Distillation

Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, Yin Cui

01 Problem, Approach, Results

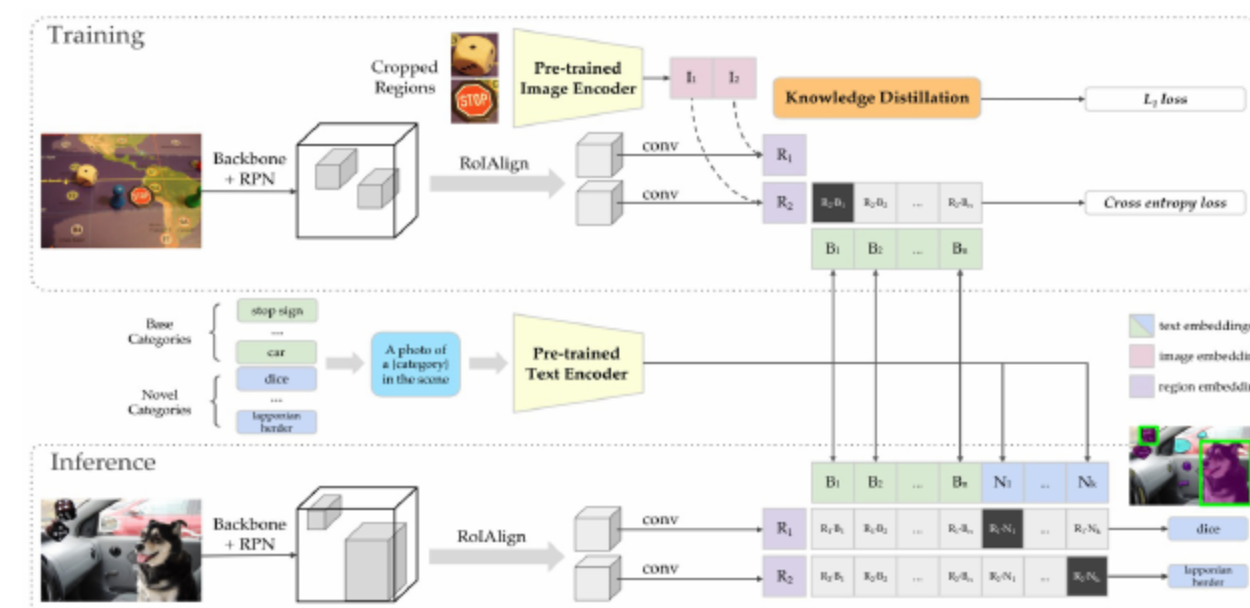


- Goal: detect novel categories using only base-category box or mask labels
- Approach: distill vision-language teachers (CLIP, ALIGN) into a two-stage detector
- Mechanisms: ViLD-text (text embeddings as classifier) and ViLD-image (visual distillation)
- Strong** novel-category AP on LVIS; zero-shot transfer to VOC, COCO, Objects365
- Scales** with stronger teachers; **efficient** compared to crop-based baselines

02 Scaling Beyond Base Categories

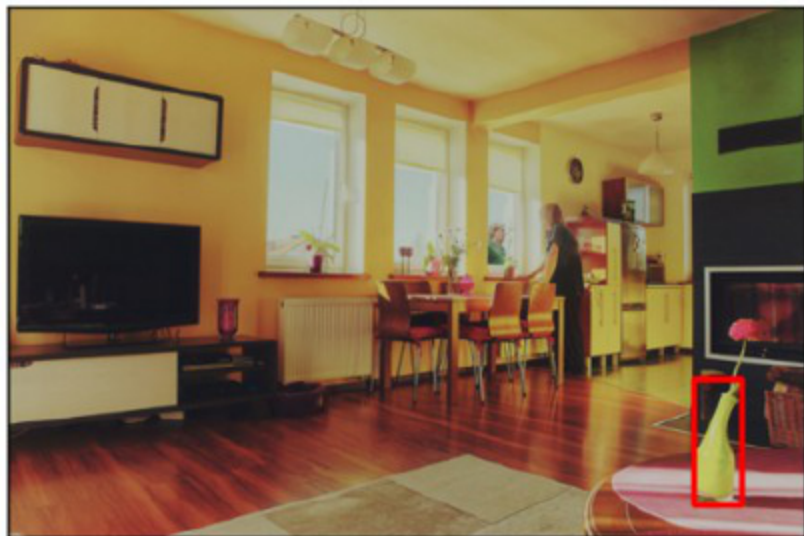
- Long-tailed datasets make full supervision expensive; rare classes have few examples
- Image-text models like CLIP and ALIGN enable **zero-shot** classification from web pairs
- Gap: transfer image-level open-vocabulary knowledge to object-level detection
- Idea**: reuse pretrained open-vocabulary classifiers to supervise detectors
- Opportunity**: expand vocabulary without collecting new detection labels

03 ViLD Pipeline



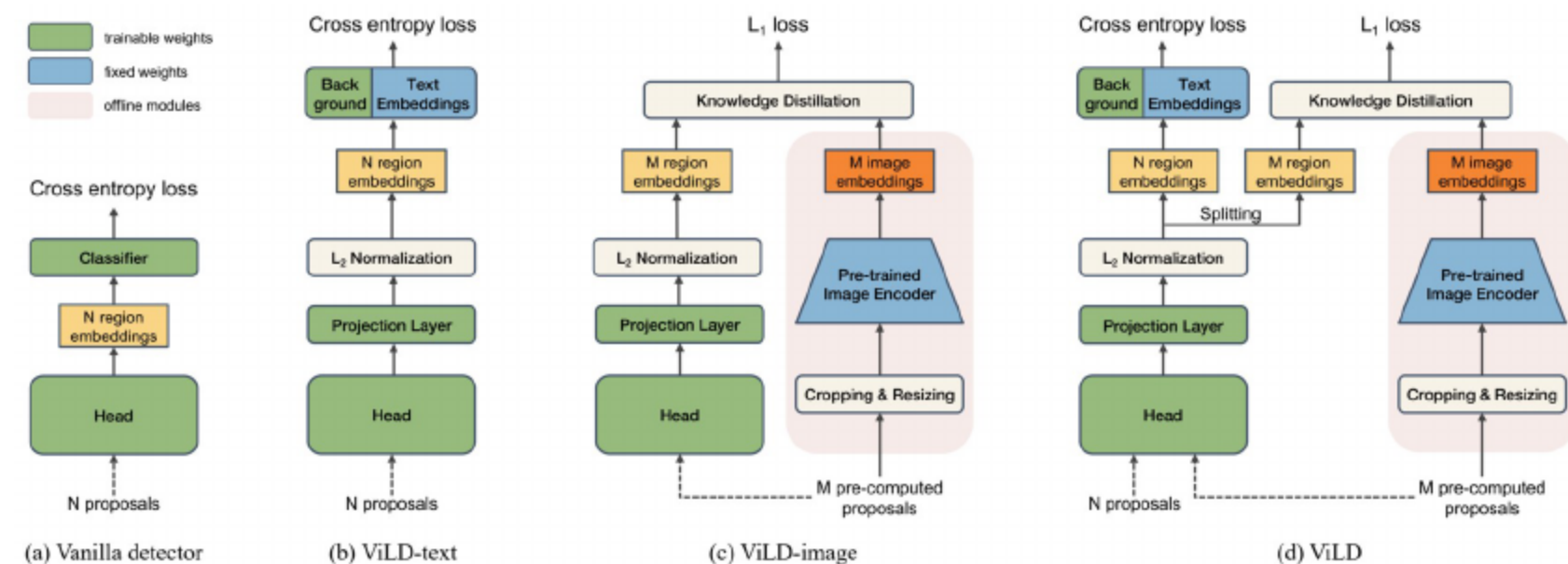
- Class-agnostic localization**: two-stage detector with class-agnostic box and mask heads
- ViLD-text: use fixed text embeddings as the region classifier instead of learned weights
- ViLD-image: align region and teacher image embeddings with an L1 distillation loss
- Inference: swap in text embeddings for novel categories for open-vocabulary detection
- Compatible** with multiple backbones and teacher families

04 Crop-Based Open-Vocab Classification



- Classify proposal crops with CLIP; combine CLIP score with objectness by geometric mean
- High** AP on novel classes but overall AP behind supervised on common and frequent
- Slow inference**: per-crop encoder calls; scores weakly reflect box quality
- Takeaway**: strong teacher, but impractical latency for deployment

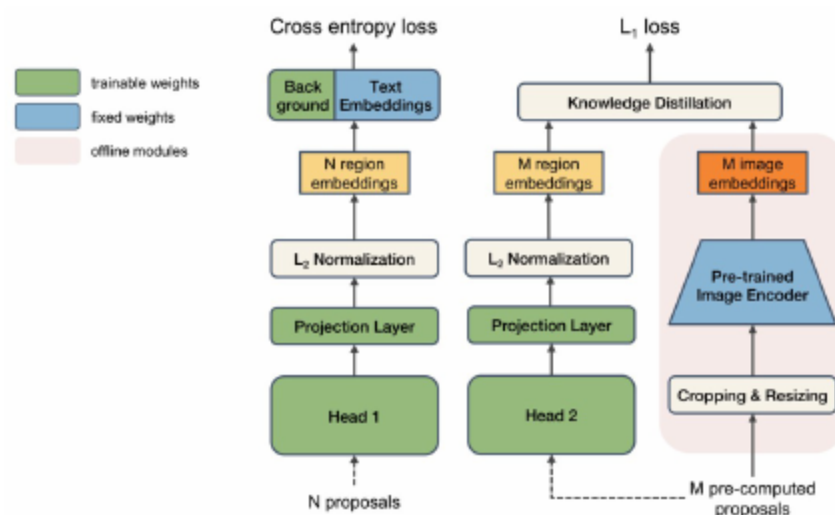
05 Replace Classifier with Text Embeddings



- Classify region features using fixed CLIP text embeddings for base categories as weights
- Add a learnable background embedding; apply softmax with a temperature
- Train end-to-end with base annotations; test by replacing or augmenting with novel texts
- Benefit**: preserves detector speed while enabling open vocabulary
- Stronger than** language-only embeddings such as GloVe on novel AP

06 Distillation + Classification

- ViLD-image: align region embeddings to teacher image embeddings on cropped proposals
- Total loss combines ViLD-text cross entropy with a weighted L1 distillation term
- Use offline proposals that can include novel objects; enables learning beyond base classes
- Option: separate heads and ensemble scores to reduce objective contention
- Result**: better novel AP than using text or image signals alone



07 RPN Generalizes to Novel

Supervision	$AR_r@100$	$AR_r@300$	$AR_r@1000$
base	39.3	48.3	55.6
base + novel	41.1	50.9	57.0

- RPN trained on base-only shows about a two point recall drop versus base plus novel
- Indicates** proposal localization largely transfers to unseen categories
- Focus on classifier transfer yields the biggest gains
- Implication**: improve heads and teacher alignment rather than proposals

Method	AP_r	AP_c	AP_f	AP
Supervised (base class only)	0.0	22.6	32.4	22.5
CLIP on cropped regions w/o objectness	13.0	10.6	6.0	9.2
CLIP on cropped regions	18.9	18.8	16.0	17.7
Supervised (base+novel)	4.1	23.5	33.2	23.9
Supervised-RFS (base+novel)	12.3	24.3	32.4	25.4

- CLIP-on-crops attains **high** novel AP and improves with objectness ensembling
- **Overall AP lags** supervised on common and frequent categories
- **Inference 630 times slower** than detector-style methods
- **Use as teacher**: great signal but not a practical runtime path

Method	AP_r	AP_c	AP_f	AP
Supervised (base class only)	0.0	22.6	32.4	22.5
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CLIP on cropped regions	18.9	18.8	16.0	17.7
Supervised (base+novel)	4.1	23.5	33.2	23.9
Supervised-RFS (base+novel)	12.3	24.3	32.4	25.4

- ViLD-text with CLIP texts **outperforms** GloVe on novel AP; better common and frequent AP than crop baseline
- ViLD-image works with no labels via distillation but trails crop-based novel AP
- **Combined ViLD boosts** novel AP; stronger teachers like ALIGN or larger ViT further raise ceilings
- Ensembling separate heads improves balance across base and novel

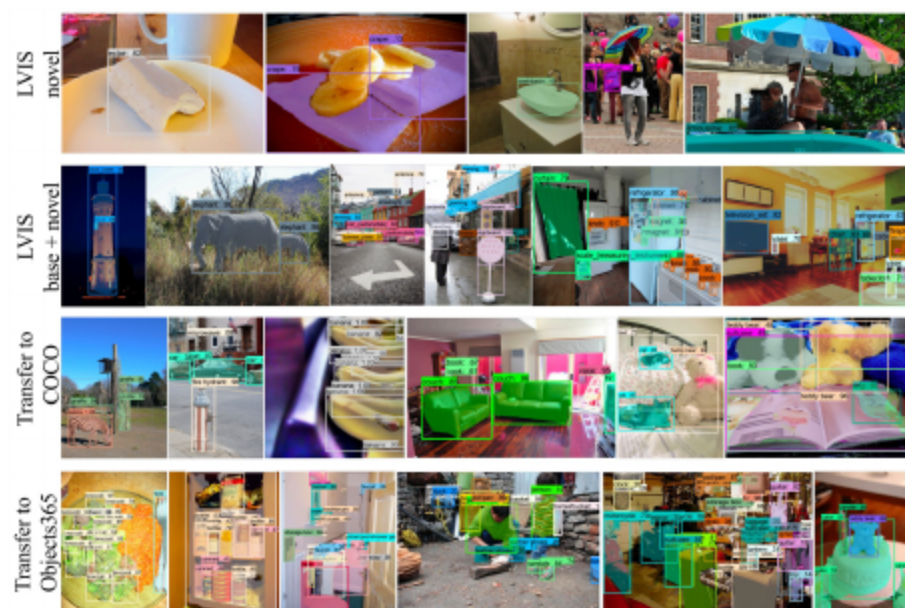
Method	Training source	Novel AP	Base AP	Overall AP
Bilen & Vedaldi (2016)	image-level labels in $C_B \cup C_N$	19.7	19.6	19.6
Ye et al. (2019)		20.3	20.1	20.1
Bansal et al. (2018)	instance-level labels in C_B	0.31	29.2	24.9
Zhu et al. (2020)		3.41	13.8	13.0
Rahman et al. (2020)		4.12	35.9	27.9
Zareian et al. (2021)	image captions in $C_B \cup C_N$ instance-level labels in C_B	22.8	46.0	39.9
CLIP on cropped regions	image-text pairs from Internet (may contain $C_B \cup C_N$)	26.3	28.3	27.8
ViLD-text		5.9	61.8	47.2
ViLD-image	instance-level labels in C_B	24.1	34.2	31.6
ViLD ($w = 0.5$)		27.6	59.5	51.3

- **Exceeds** prior open-vocabulary and zero-shot detectors on novel and base under the COCO protocol
- Uses off-the-shelf classifier as teacher; no detection-tailored pretraining
- Distillation leverages proposals that include novel objects
- **Stronger generalization** than text-only heads when base set is small

Method	PASCAL VOC [†]		AP	COCO		AP	Objects365	
	AP_{50}	AP_{75}		AP_{50}	AP_{75}		AP_{50}	AP_{75}
ViLD-text	40.5	31.6	28.8	43.4	31.4	10.4	15.8	11.1
ViLD	72.2	56.7	36.6	55.6	39.8	11.8	18.2	12.6
Finetuning	78.9	60.3	39.1	59.8	42.4	15.2	23.9	16.2
Supervised	78.5	49.0	46.5	67.6	50.9	25.6	38.6	28.0

- Swap classifier texts to transfer an LVIS-trained detector to VOC, COCO, Objects365
- ViLD **outperforms** ViLD-text in transfer; gaps remain versus supervised or finetuned
- **Plug-and-play** deployment via language without finetuning
- Background embedding reused from LVIS works adequately

12 Capabilities and Interactive Detection



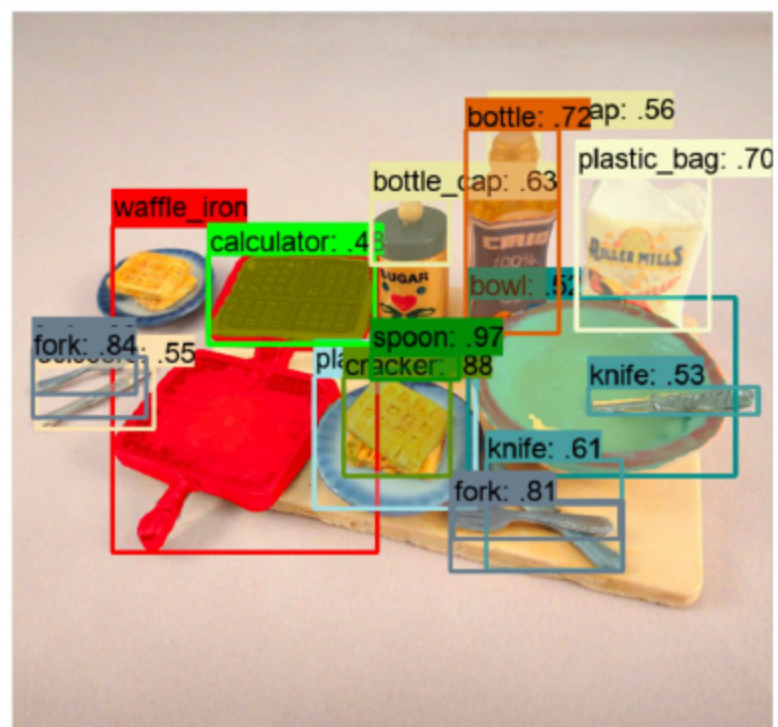
- **Detects** novel categories and transfers across datasets without finetuning
- Interactive queries: attributes and fine-grained labels via on-the-fly text prompts
- Maintains base-category performance while adding open-vocabulary capability
- **Example uses:** colors, fine-grained species, attribute-conditioned search

13 Attributes and Fine-Grained



- Systematically expand categories with attributes to form combinations like colors times fruits
- Compose fine-grained species with text prompts at inference
- Conditional scoring akin to YOLO9000 enables scalable vocabulary growth
- **Practical path** to richer detectors without extra annotations

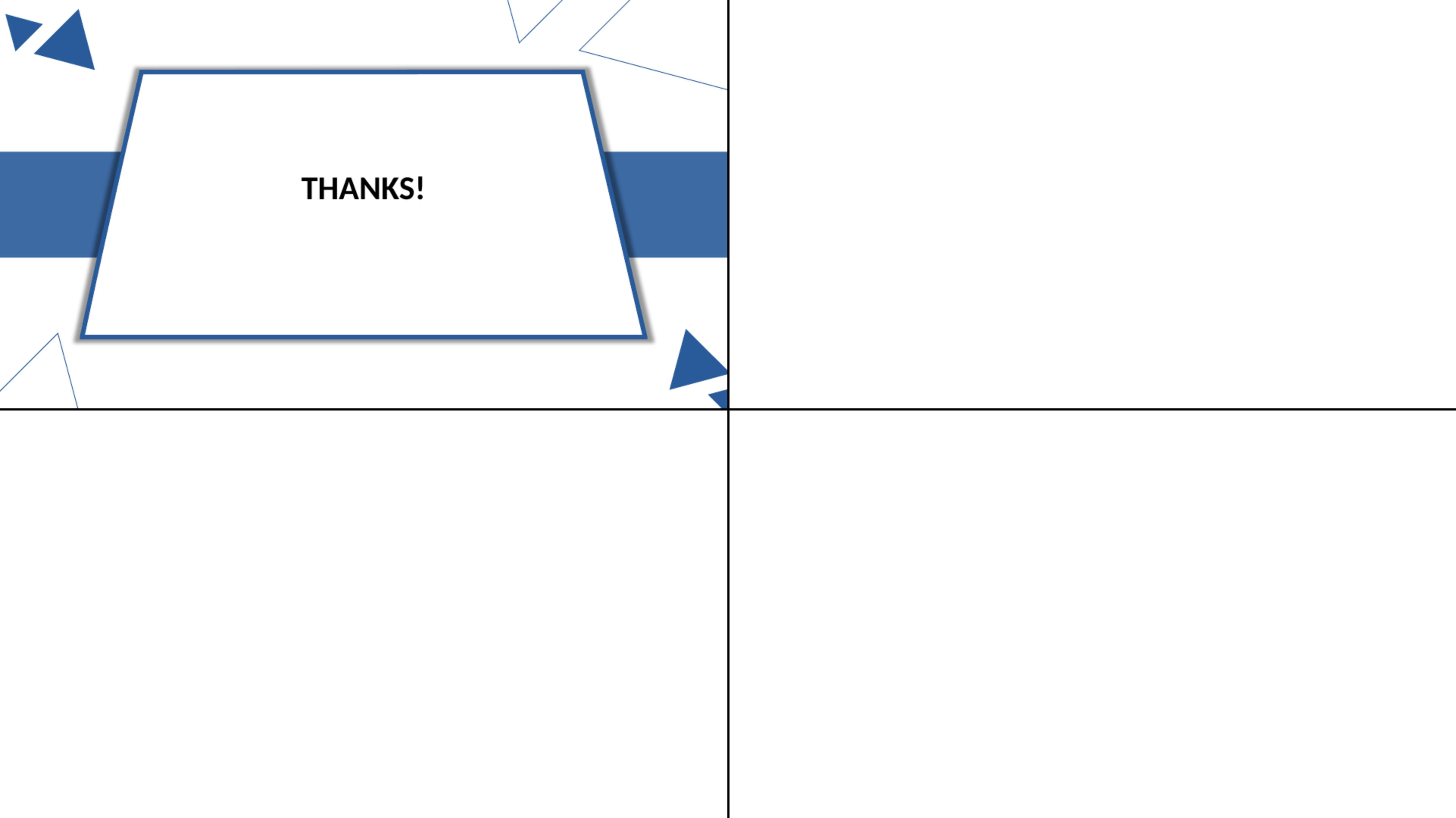
14 Failure Modes



- **Missed detections** on novel objects; confusion among visually similar classes
- **CLIP-on-crop sensitivity** to aspect ratio and context within resized crops
- **Class-agnostic masks** may rely on low-level cues, yielding incomplete masks
- **Open issues:** pose understanding and very fine-grained distinctions

15 Key Takeaways

- Distill CLIP and ALIGN into two-stage detectors for efficient open-vocabulary detection
- **Strong** LVIS novel AP; performance scales with stronger teachers
- Zero-shot transfer by swapping text embeddings across datasets
- **Simple, scalable** alternative to annotation-heavy vocabulary expansion

The background features a white canvas with a thin black horizontal line across the middle and a thin black vertical line on the right side. In the upper-left quadrant, there are several blue geometric elements: a small triangle, a larger triangle, a horizontal rectangle, and a parallelogram. In the upper-right quadrant, there are thin blue lines forming a 'V' shape and a larger triangle. In the lower-left quadrant, there is a thin blue line forming a triangle. In the lower-right quadrant, there is a blue triangle pointing towards the center.

THANKS!